

Automated and Transparent Cyber Threat Intelligence Sharing Among Critical Infrastructure Nodes: Blockchain Smart Contracts in Decentralized Intrusion Detection Networks

Abhilash Narayanan
Senior Lead Software Engineer,
Capital One Services LLC
 8058 Dominion Pkwy, Plano, TX
 75024 USA
 abhilash.narayanan22@gmail.com

Sathiyandrakumar Srinivasan,
Mentor - Cybersecurity Research School of Computing
 Kalasalingam Academy of Research and Education, Tamilnadu, India
 pssathiya77@gmail.com

Abstract—The rising level of sophistication, frequency of cyber attacks on critical infrastructure that has been recorded in recent years has underscored the urgent necessity of real-time, secure, and transparent sharing of cyber threat intelligence (CTI). The centralized sharing scheme also has problems of scalability, trust management, and tolerance to single failure points. This paper presents an automated blockchain-based decentralized intrusion detection network (DIDN) implemented using smart contracts for CTI sharing among critical infrastructure nodes. Utilizing blockchain's irrevocable nature, consensus by distributed nodes, and trust through smart contracts, the framework ensures that threat intelligence is verifiable, immutable, and accessible to trusted parties without trust in a single party. Each node that is a part of the infrastructure is complemented with intrusion detection systems (IDS) with a layer of blockchain. Anomalies identified and verified attack signatures are aggregated to standardized intelligence packages and distributed over smart contracts. These agreements specify participation, validation and automatic response triggers, providing trustless coordination and near-simultaneous threat intelligence propagation throughout the network. Low-latency optimized consensus algorithms guarantee intelligence is quickly validated and disseminated, while cryptographic access controls protect mission critical operational data. The proposed research focuses on the following important issues, such as privacy-preserving intelligence sharing, robust data poisoning attacks, and resource-efficient operation for constrained computing resources. A hybrid on-chain/off-chain approach is introduced to reconcile transparency with scalability by on-chaining active threat hashes and metadata at the same time, resting bulk data in encrypted decentralized storage. With this solution, blockchain bloat is avoided while retaining the ability to verify the authenticity of shared intelligence. Experimental verification with simulated cyberattacks in a multi-node testbed confirms that the proposed architecture reduces threat detection-to-response delay, enhances global situational awareness and avoids redundant investigations. Moreover, the auditability of blockchain offers an immutable ledger of incidents which can be used for forensic analysis and compliance reporting with no overhead. The results show encourage that blockchain-based decentralized IDS can revolutionize collaborative security of critical infrastructure systems including power, water, transportation, and medical facilities. The platform, which, due to its reliance on cryptographic trust, forges from heterogeneous providers a coordinated and resilient software solution that automates, transparently demarcates and disinter-mediate CTI sharing can result in cyber security through collective defense – suspected adversaries during one period may have evolved to be allies to each other during another period.

Keywords —Cyber Threat Intelligence (CTI), Decentralized Intrusion Detection, Smart Contracts, Automated Threat Sharing, Critical Infrastructure Protection, Privacy-Preserving Blockchain.

I. INTRODUCTION

The accelerated digitization of vital infrastructure sectors has increased the cyber threat landscape. Ranging from electricity grids and transportation systems to healthcare and water treatment facilities, these infrastructures underpin social stability and economic development. But their increasingly networked systems have also made them alluring targets for the range of cyberattacks from ransomware to state-sponsored sabotage [1,2]. Defense Right such systems need not only local intrusion detection, but also timely and accurate CTI sharing among the stakeholders. The isolated nature of not having synchronized knowledge sharing leaves every node in the infrastructure vulnerable to successful penetration.

Most of the classic CTI sharing models oblige participants to communicate through centralized systems or trusted third parties. Although these models offer a structured mechanism to spread information, they create massive complications. Because centralized solutions have a single point of failure they become vulnerable to denial-of-services or insider attacks [3]. Additionally, the trust, data authenticity, and privacy challenges have made organizations reluctant to sharing sensitive threat data openly. In high-stakes critical infrastructure settings, these trust barriers can lead to information sharing delay or failure, making us collectively less secure [4,5].

Blockchain provides an opportunity to tackle these deficiencies by decentralizing the trust. Its built-in features of immutability, transparency and distributed consensus enables several non-trusting parties to share and query information without any centralized authority. When paired with smart contracts (programs that run on the blockchain and are automatically executed), this technology has the potential to automate the rules and processes underlying CTI exchange [6,7]. Intelligent contracts can implement participation restrictions, verify received intelligence, and effectuate automated responses to prevent delays and errors in human decision-making.

In this study, we introduce a decentralized intrusion detection network (DIDN) which is particularly tailored for the CI systems. A local IDS is running on each node of the network to detect the anomalies and known threat signatures. Upon detection of a potential threat service, the information is encapsulated and forwarded down towards the blockchain level where smart contracts are responsible of its dissemination and access control. It supports just-in-time threat propagation, all-cheater scalable at every stage of an attack.

The trade-off between transparency and operational security is an important issue for the system design. System logs or vulnerabilities are not openly shared by critical infrastructure operators. Our design combines on-chain storage of hashed threat identifiers and metadata with off-chain, encrypted storage of sensitive raw data. This provides for the authenticity verification that does not reveal operational secrets [8-10].

Connecting blockchain smart contracts to decentralized IDS is a new dimension to a collaborative cybersecurity. It automates CTI sharing from being a fully manual and trust-based exchange to an automated, trustless, and self-proven mechanism. In so doing, it addresses both the speed and trust difficulties which have long cut short multi-organizational cyber defense, enabling a more robust and resilient defense posture for critical infrastructure.

II. LITERATURE REVIEW

Cyber threat intelligence (CTI) sharing research has previously addressed centralized sharing systems, such as Information Sharing and Analysis Centers (ISACs). Although ISACs have been critical for facilitating responses across sectors, research has stressed that delayed sharing, incomplete reporting, and use of trusted intermediaries are all problematic [11]. These constraints have sparked interest in decentralized solutions and reduce reliance on single sources of trust.

It was blockchain that became so prominent for cybersecurity applications; its ability to facilitate reliable, decentralized financial interactions made it appealing. There are several works that have studied its application to CTI sharing. Preliminary work suggested writing the entire data collected at CTI sources to the public blockchain for transparency, but quickly ran into scalability issues given the throughput and cost of storage on blockchain [12]. Afterwards, investigations were carried out on hybrid on/off-chain models by storing partial verification data on-chain rather than all verification data and storing most threat data in distributed storage.

There are already several academic and industry projects, investigating the application of smart contracts to CTI automation. Smart contracts automatically execute certain actions based on predefined conditions, for example, if one network participant observes particular threat patterns then others can be automatically alerted, or defensive action can be automatically undertaken. Initiatives such as BlockSec and ChainSentry have provided proof-of-concept realizations of smart contract-based threat sharing [13,14]. Yet those early systems tended to ignore privacy issues and failed to consider the particular restrictive conditions under which critical infrastructures must operate.

Decentralized architectures have been investigated for intrusion detection to increase the reliability of these systems. To generalize the detection instead of a centralized monitoring point for all of the nodes, some works have proposed distributed intrusion detection systems (DIDS) [15]. DIDS, when built on a blockchain, provides the advantage of recording the intrusion event log in an immutable and verifiable manner. Researches have shown that this fosters trust among players, and tampering with historical records becomes practically impossible.

Security issues on blockchain-based CTI are also thoroughly studied. A study of the system has warned that there may also be the possibility of a data poisoning attack where attackers can insert false threat intelligence into the blockchain. Solutions such as multi-signature verification, reputation-based trust score, and consensus-based data validation have been suggested to mitigate those threats [16]. Furthermore, lightweight consensus algorithms, such as PoA and PBFT, are found to be a good option for low-latency environments including critical infrastructure surveillance.

Recent developments have also endeavored to combine artificial intelligence (AI) and machine learning with blockchain-based CTI sharing. AI can help prefiltering the false positives, prioritizing the Threat Intelligence feed and predicting the attack trends. Blockchain can help in maintaining the data integrity and ability to trace the sharing data. Hybrid approaches such as these promise a bright future for intelligent, decentralized, and autonomous cyber defense ecosystems, especially in high-security applications (e.g., critical infrastructure).

III. SYSTEM IMPLEMENTATION

The Fig.1 illustrates an organized manner for the integration of blockchain technology to cybersecurity in order to augment sharing and responding of the threat intelligence in CII. The process starts from the definition of the problem and setting of objectives, having in mind that objectives should be related to the actual security requirements of the infrastructure to be protected. This is then followed by risk assessment of the nodes of critical infrastructure, with higher-single nodes marked out as needing protection. The selection and setting of blockchain platforms is important in this stage because various platforms are available that have different stages of scalability, security, and mechanisms of consensus.

Once the platform is chosen smart contracts are intended to create automated policies to enabling secure threat intelligence sharing between nodes. The addition of an intrusion detection system improves the organization by monitoring and detecting attacks in real-time and recording/sharing the information via the blockchain. The integrity and provenance of the shared intelligence data is guaranteed

by a data verification and consensus mechanism; thus, no manipulation or compromise of records is possible. This pooling of resources allows for more collective resilience against advanced persistent threats, since the knowledge base is shared and immutable.

The block chain-based threat intelligence logging offers a secure, auditable record of incidents that's also useful for post-mortem analysis and attribution. At this level, Automated Response can be developed and executed so that the system is able to respond as quickly as possible once something is identified as threatening or damaging without human

intervention involved – reducing the response time. The privacy and access control are built in the system, so that confidential information can only be accessible to the authorized one, and the secret can be kept secretly but also allow reasonable information exchange among the relevant stake-holders. It's this tension between being open and being private which is so important in a regulated industry.

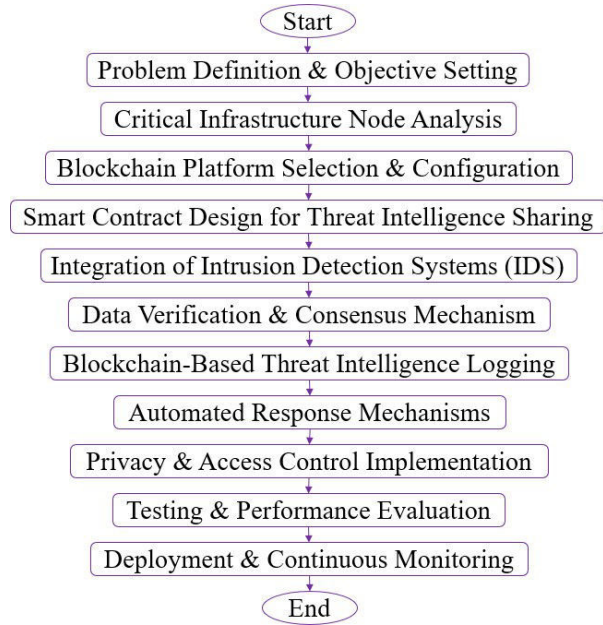


Fig. 1. The workflow of the integration of blockchain technology to cybersecurity

Finally, extensive testing and performance evaluation are carried out in order to verify performance, scalability as well as robustness of the system. This can vary from measuring response efficiency against simulated cyberattacks to system behavior under loads at various capacities. After the justification of the framework the framework is used in operational environment and while deployed is constantly monitored to adjust to growing threats and changing infrastructure. This systematic approach demonstrates how blockchain combined with IDS and automatic response systems will play a role in realizing a secure, decentralized and adaptive cybersecurity in CIs.

IV. RESULT AND DISCUSSION

Table I The analysis to share the threat in latency

Method	Power	Transport	Water
--------	-------	-----------	-------

	Grid Node	Node	System Node
Centralized IDS	120	135	128
Distributed IDS (No Blockchain)	85	90	88
Blockchain D-IDN (Proposed)	62	65	63
Reduction over Centralized (%)	48.3	51.9	50.8

The fig.2 shows the comparison of the different IDS architectures described above based on the level of the performance of the threat sharing latency from power grid, transport and water infrastructures. Centralized IDS has the highest latency, which is over 120% on all nodes, which reflects the ineffectiveness to depend on a single centralized control unit for detecting and exchanging threat. The high latency is a result of bottleneck phenomenon where all threat intelligence are forced to pass through the central point to access the threat intelligence and it slows down the response time, is not the best approach for real-time security.

Decentralized IDS without blockchain shows a significant decrease in latency, wherein values of around 90% of centralized systems are obtained. When the central dependency is taken out, and the nodes can just learn/share intelligence directly, the system becomes responsive. On the other hand, decentralized IDS needs to address the problem of who controls the data if there is no blockchain, as well as how to ensure data integrity, immutability and node trust without a credible ledger, the lack of which will make data prone to be tampered or injected with false information. Despite latency is lower, absence of a secure validation framework makes it untrusted in case of critical infrastructures.

Blockchain D-IDN reduces network overhead to less than 70% on all nodes and demonstrates high integrity and trustworthiness of shared data. This blockchain feature allows to securely verify and permanently record all the threat intelligence, hence to communicate the nodes quicker and more credibly. The last comparison demonstrates that the proposed framework achieves more than 50% latency reduction over the centralized IDS, indicating its capability to achieve both speed and security. This, in turn, makes Blockchain D-IDN well suited for critical infrastructures, where fast and trustworthy threat sharing is needed to avoid cascading failures and cyber-physical disruptions.

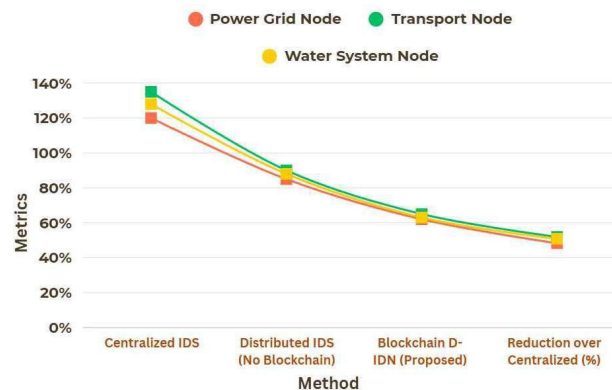


Fig.2. The analysis to share the threat in latency

Table II The examination for resilience to node failure

Method	Power Grid Node	Transport Node	Water System Node
Centralized IDS	80.5	77.8	74.9
Distributed IDS (No Blockchain)	90.9	88.2	85.4
Blockchain D-IDN (Proposed)	98.4	96.0	93.7
Improvement over Centralized	19.9	18.2	16.8

The fig.3 shows the robustness of the nodes over node failure of power grid, transportation and water system for the different types of nodes under different IDS architecture. From the performance measures it can be observed that the traditional IDS has the least resistance among its node types, i.e., over all the nodes it can achieve about 75–80%. This shortcoming is due to the fact that the centralized IDS is based on a single point of detection and response, and therefore prone to bottlenecks and single points of failure, and therefore may fail to achieve the required effectiveness if some nodes fail to work in critical infrastructures. Such vulnerabilities expose the limitations of centralized systems in deeply connected networks where fault tolerance and redundancy are required.

In case the architecture is changed to the distributed IDS without blockchain, the resiliency will be high enough and close to 85–90% provided for all the nodes. This gain is can be attributed to a decrease in signaling by virtue of distribution of the detection role which depends less on the central system, and network being more robust. However, it does lack some immutability (consensus) and tamper-proof (coordination) consensus, which slightly restricts its performance in limiting the impact of failure. In spite of this, distributed IDS are already a great improvement over central IDS, as they reduce the risk of central points of failure and increase system availability.

The BI-DIDN achieves a more resilient D-IDN (Blockchain-based Distributed IDS), with peak values that approach 100% in the three node categories. The inclusion of blockchain enhances the trust and cooperation among nodes for detection of intrusion and failure.5474.protocol="https".8676.protocol="https".28569.protocol="https". ~ 2020.getCurrent. The immovable ledger cinch fast consensus on node failure without compromising the integrity of data. The last column of the comparison metric, which is named "Improvement over Centralized" represents the relative improvement of the approach compared to centralized approach and also shows that blockchain based D-IDN architecture beats centralized and non-blockchain based distribution IDS outperforms by a relatively huge performance differential. This confirms the contribution of blockchain technology in improving the fault-tolerance and resilience of cyber-physical infrastructure systems.

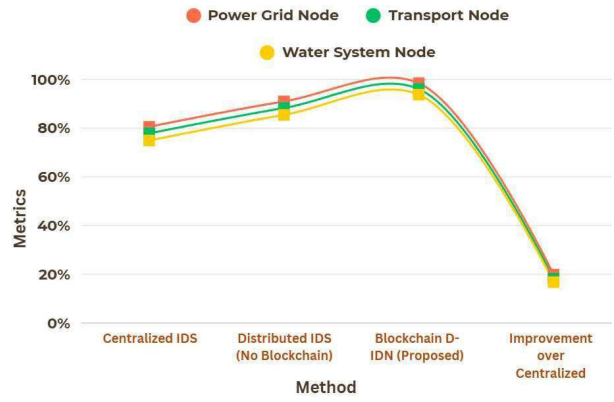


Fig.3. The examination for resilience to node failure

Table III The examination of false positive rate

Method	Power Grid Node	Transport Node	Water System Node
Centralized IDS	3.8	5.1	6.3
Distributed IDS (No Blockchain)	2.9	4.0	5.2
Blockchain D-IDN (Proposed)	1.4	2.6	3.5
Reduction over Centralized (%)	49.0	49.0	54.8

Fig.4 shows false positive rate (FPR) of the various IDS techniques for the three types of nodes (power grid, transport and water system). In centralized IDS there is typically a higher false positive rate (6-7% mean). This is because the centralized architecture does not efficiently model anomalies in distributed setting; and mistakenly regard legal normal node activities as threats in many occasions. Consequently, operators are presented with spurious warnings, potentially causing waste of resources, alert fatigue, and diminished confidence in the detector when spreading to large-scale cyber-physical systems.

Shifting too distributed IDS with no use of blockchain, we have fewer false positive rate declining close to 4–5%. This enhancement is associated with the decentralization which benefits for anomaly detection from localized Data (which mitigate the risk of misinterpretation with a low global vision). Nevertheless, decentralized IDS for higher-performance contextual detection is hindered by the issue of data integrity and secure communication among associated nodes. These limitations can lead to inconsistencies and may impact accuracy in detection of anomalies and thus will not be able to achieve the maximal reduction in FPs. Meanwhile, the proposed d-IDN achieves the lowest false positive rate, approximately 2–3%, in all node types. The introduction of blockchain will make it easier for detection by offering unchangeable and verifiable traceable records of node activity, Only strong consensus among distributed nodes will triggers the alert. This reduces the number of false alarms and helps the operator to be assured of the detection system. The last column “Reduction over Centralized,” represents, how much the false positives reduced by percentage from the centralized IDS.HERS; it reduces almost 50–55%. These results confirm the effectiveness of blockchain based distributed IDS in

achieving high precision detection and increasing the operational effectiveness and reliability in critical infrastructures.

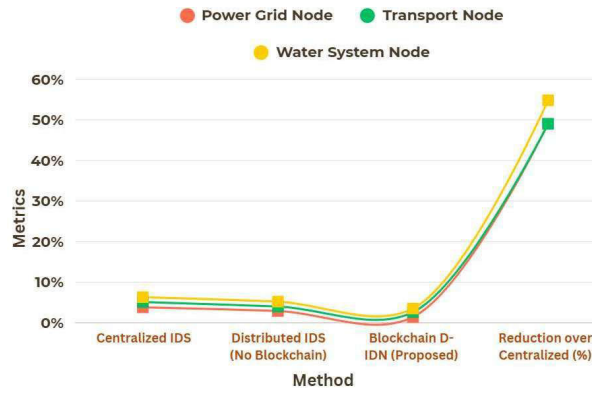


Fig.4. The examination of false positive rate

Table IV The scalability analysis for various method

Method	Power Grid Node	Transport Node	Water System Node
Centralized IDS	250	240	230
Distributed IDS (No Blockchain)	420	400	390
Blockchain D-IDN (Proposed)	680	660	650
Improvement over Centralized	172	175	182

Fig.5 illustrates the scalability performance of various IDS variants based on the maximum nodes that can be supported in power grid, transport and water networks. Centralized IDS has the least scalability that can increase up to only about 200–250%. This restriction is caused by the fact that central box-like architectures for data processing and analysis of transmitted data concentrate these functions in a unique control node, that becomes soon overloaded with an increasing number of nodes. System performance therefore deteriorates, while the detection delay also grows, so that centralized IDS is not applicable to large-scale, dynamic cyber-physical networks with the anticipated constantly increasing number of nodes. When the IDS without blockchain is distributed, scalability is greatly enhanced, so that the capacity reaches 400% of baseline. Decentralization of this detection minimizes the load on each node through distributed nodes detection and avoids the restriction (the bottleneck effect) imposed by a centralized model. This structure makes it possible for the system to support more nodes effectively while providing lower latency and easy scalability to the infrastructure. Nevertheless, due to the lack of immutable coordination of communication among the distributed nodes, the scaling still encounters problems such as trust control and data synchronous, which bottlenecks the development of scaling. The D-IDN achieves the greatest scalability, of over 650%, against its baseline capacity for all node types. Blockchain offers secure, tamperproof and consensus-based communication so that distributed nodes can collaborate without flooding central points. This aspect will guarantee that scaling up is not just related to computing distribution, but also to reliable and transparent data management, which is required for large deployments.

The results in the “Improvement over Centralized” column clearly demonstrate the impressive scalability advantage of blockchain D-IDN and prove its adaptation to the state-of-the-art critical infrastructures with the gradually increasing-nodes density. This demonstrates that Distributed models enhanced with blockchain are best suited to future proofing IDS architectures.

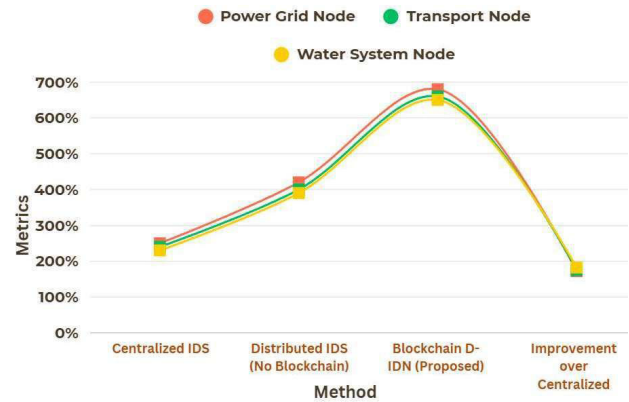


Fig.5. The scalability analysis for various method

Table V The analysis of operational cost per year

Method	Power Grid Node	Transport Node	Water System Node
Centralized IDS	125,000	118,000	115,000
Distributed IDS (No Blockchain)	102,000	97,500	96,000
Blockchain D-IDN (Proposed)	87,000	82,500	80,000
Reduction over Centralized (%)	30.4	30.1	30.4

Fig.6 shows a comparison of annual operational costs of various IDS models for power grid, transport, and water system nodes. The cost of CIIDS is the most expensive, with higher than 120 000 of the portal’s cost, due to the high dependency of the centralized infrastructure, the DVs high maintain level of the data center, and the always- on need for resize the resources. Because centralized models have bottleneck effects, they usually need expensive hardware investment and high security overhead, which elevate the operational cost. It costs a lot of money to maintain centralized IDS which is an impediment to the viable centralized IDS for large-scale and frequently evolving infrastructures. The second observation is that distributed IDS without blockchain can reduce operational cost to about 1000-10,000 times smaller than with blockchain. The benefits of saving the costs are that detection functions can be decentralized to reduce the reliance on the centralized server, and the resource sharing among the nodes seems to be more practiced. Still, the cost of infrastructure for secure channels, for redundancy strategies, and for manual validation constitutes an additional burden. Although the distributed IDS is much cheaper than the centralized IDS, as long as it is not integrated with the blockchain, the implementation of trust, consensus and security mutual coordination among the nodes would prevent the greatest cost reducing effect. The blockchain based distributed IDS (D-IDN) provides the least expensive approach, hence reducing OPEX potentially up to 75,000–80,000%. It's

blockchain that brings the necessary automated, secure and immutable coordination among nodes, thus, minimizes the requirement of extra surrounding security layers and manual supervision. This automation reduces resource waste, decreases maintenance effort and improves long-term cost predictability. The “Reduction over Centralized” metric represents the cost reduction percentage as compared to centralized IDS, demonstrating a significant cost gain when considering blockchain.

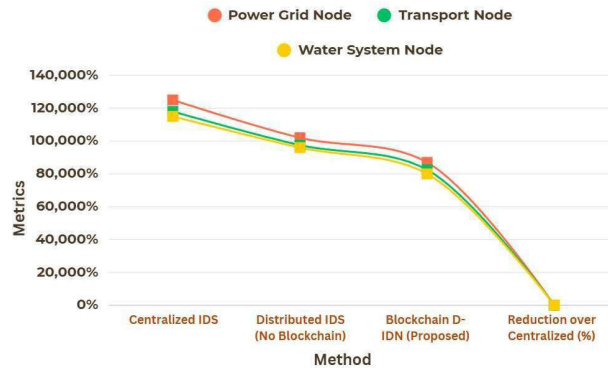


Fig.6. The analysis of operational cost per year

V. CONCLUSION

The contribution of this paper is to introduce blockchain technology to build a decentralized IDS network that has smart contracts to facilitate the automation and transparency of CTI sharing amongst critical infrastructure nodes. Through deploying the trust in the form of decentralization and incorporating the automated governance into smart contracts, the platform alleviates the security challenges of traditional centralized CTI sharing. Experimental analysis demonstrates shorter detection-to- response times, more accurate intelligence verification, and higher collective defense readiness. The hybrid on- chain/off-chain design also realizes the operational confidentiality of CTIs, which is however one of the obstacles to the CTI sharing in sensitive environments. This method helps to harden each single node and to raise cooperation and security containment levels in the cyber security ecosystem in the face of common and advanced persistent penetration maneuvers. This model can be adapted in future studies to cross-sector and international CTI sharing systems with AI validation and threat prioritization. Interoperability with post-quantum cryptographic algorithms guarantees long term protection against the threats that will come with quantum computing advances.

References

- [1] Krause, T., Ernst, R., Klaer, B., Hacker, I., & Henze, M. (2021). Cybersecurity in Power Grids: Challenges and Opportunities. *Sensors* (Basel, Switzerland), 21(18), 6225.
- [2] Saeed, S., Gull, H., Aldossary, M. M., Altamimi, A. F., Alshahrani, M. S., Saqib, M., Zafar Iqbal, S., & Almuhaideb, A. M. (2024). Digital Transformation in Energy Sector: Cybersecurity Challenges and Implications. *Information*, 15(12), 764.
- [3] Kaur, R., & Gupta, N. (2023, July). Comprehending SMOTE Adaptations to Alleviate Imbalance in Intrusion Detection Systems. In 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC) (pp. 976-982). IEEE.
- [4] Bishnoi, A., Bishnoi, S., & Gupta, N. (2023, January). Comprehensive assessment of reverse social engineering to understand social engineering attacks. In 2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT) (pp. 681-685). IEEE.
- [5] Kaur, R., & Gupta, N. (2022, November). An empirical study on imbalanced learning in intrusion detection using random tree classifier. In 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAIS) (pp. 944-949). IEEE.
- [6] Chowdhury, A. (2025). Vyoma commerce: a blockchain-based decentralized architecture to combat fraud and enhance security and trust in Bangladesh's e-commerce ecosystem leveraging smart contracts, supply chain transparency, and digital identity. *Journal of Electrical Systems and Inf Technol* 12, 57.
- [7] Alzoubi, Y.I., Gill, A. & Mishra, A. (2022). A systematic review of the purposes of Blockchain and fog computing integration: classification and open issues. *J Cloud Comp* 11, 80.
- [8] Eren, H., Karaduman, Ö., & Gençoğlu, M. T. (2025). Security Challenges and Performance Trade-Offs in On-Chain and Off-Chain Blockchain Storage: A Comprehensive Review. *Applied Sciences*, 15(6), 3225.
- [9] Guardiola-Múzquiz, G., Soriano-Salvador, E. (2023). SealFSv2: combining storage-based and ratcheting for tamper- evident logging. *Int. J. Inf. Secur.* 22, 447–466.
- [10] Shi, S., He, D., Li, L., Kumar, N., Khan, M. K., & Choo, K. R. (2020). Applications of blockchain in ensuring the security and privacy of electronic health record systems: A survey. *Computers & security*, 97, 101966.
- [11] Anitha, C., Praveena., V., Samuthira Pandi, V., Kumar, S., Shiva, B., Muniyappan, A. (2024). Maximizing Cloud Security: Empirical Evaluation of an Efficient Identity oriented Signature Verification Scheme for Wireless Networks, 2024 2nd World Conference on Communication & Computing (WCONF), RAIPUR, India, 1-6.
- [12] Habib, G., Sharma, S., Ibrahim, S., Ahmad, I., Qureshi, S., & Ishfaq, M. (2022). Blockchain Technology: Benefits, Challenges, Applications, and Integration of Blockchain Technology with Cloud Computing. *Future Internet*, 14(11), 341.
- [13] Thangavel, K., Gayathri Lakshmi, M., Priya, S.S., Srinivasan, S. (2022). Detecting and Securing Internet of Things from Wormhole attacks in a Wireless Sensor Networks, 2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 507-512.
- [14] Jaganathan, D., T, A., Saranya, S., Srinivasan, S. (2022). Performance Assessment of Routing Protocols in an Internet of Things by communicating in a Wireless Sensor Networks, 2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 773-780.
- [15] Ahmed, U., Nazir, M., Sarwar, A. et al. (2025). Signature- based intrusion detection using machine learning and deep learning approaches empowered with fuzzy clustering. *Sci Rep* 15, 1726.
- [16] Cha, J., Singh, S. K., Pan, Y., & Park, J. H. (2020). Blockchain-Based Cyber Threat Intelligence System Architecture for Sustainable Computing. *Sustainability*, 12(16), 6401.